Chapter 11

Managing Weight and Eating Behaviors

1. Maintaining a Healthy Weight
2. Body Image and Eating Disorders
3. Lifelong Nutrition

Lesson 1 — Maintaining a Healthy Weight

The Calorie Connection

Metabolism is the process by which your body breaks down food and converts it into energy. Weight is a reflection of your energy balance:

Calories in > out	Weight gain
Calories out > in	Weight loss
Calories in = out	Weight stays the same

It takes about 3,500 calories to equal 1 pound of body fat. Cutting 500 calories/day through eating less, moving more, or both can result in losing about 1 pound per week. Slow, steady, and sustainable always beats fast and drastic.

Body Mass Index and Healthy Weight

Body Mass Index (BMI) measures body weight relative to height as a screening tool. It isn't perfect — a muscular athlete may have a high BMI with very little body fat. Your doctor can interpret your BMI using age- and sex-specific growth charts.

Underweight	Below standard range — may lack energy, tire easily, miss critical growth nutrients

Healthy weight	Within standard range for height — associated with best long-term health outcomes
Overweight	Heavier than standard range — current data: nearly 1 in 3 U.S. youth
Obese	Excess body fat that carries significant health risks — affects 21.1% of teens ages 2–19

Managing Your Weight — Healthy Habits

- **Realistic goals** — Aim for a consistently healthy diet and regular physical activity, not rapid weight loss
- **Nutrient-dense foods** — Fruits, vegetables, and whole grains give you more nutrition per calorie
- **Watch portions** — Restaurant and packaged portions have grown dramatically — they're often 2–3x a normal serving
- **Limit added sugars** — They add calories without meaningful nutrition
- **Stay active** — 60 minutes of physical activity on most days
- **Stay hydrated** — Girls need ~9 cups/day; boys ~13 cups

Lesson 2 — Body Image and Eating Disorders

Your Body Image

Body image is the way you see and feel about your body. Two of the biggest drivers of negative body image today:

Social media	Instagram and TikTok flood feeds with heavily filtered, edited images of 'ideal' bodies. Research consistently links high social media use with negative body image, especially in teen girls.
Peer comparisons	Comments from peers, teasing about weight, or comparing yourself to others can chip away at how you feel about your body.

The truth: healthy bodies come in all shapes and sizes. You cannot change your basic body type, and attempting to force your body into a shape it wasn't built for can seriously harm your health.

Fad Diets

Fad diets promise quick results but aren't based on sound science. They tend to be trendy for a short time, then fade — because they don't work long-term. **Weight cycling** (losing and regaining repeatedly) is harder on your body than maintaining a stable weight. Any plan that doesn't align with MyPlate guidelines is likely depriving your body of nutrients it needs.

Eating Disorders

Eating disorders are extreme, harmful eating behaviors classified as mental illnesses. About **9% of Americans** — 28.8 million people — will have one in their lifetime. They affect all genders, body sizes, races, and backgrounds. One in three people with an eating disorder is male. Average age of onset: 12–13 years old.

Anorexia Nervosa	Intense, irrational fear of weight gain leads to severe food restriction and essentially starvation. People often have a distorted body image — seeing themselves as overweight even when dangerously thin. One of the highest mortality rates of any mental illness.
Bulimia Nervosa	Cycles of binge eating followed by purging (vomiting, laxatives, or excessive exercise). Often within normal weight range — making it harder to detect. Causes severe dehydration, tooth erosion, and dangerous electrolyte imbalances.
Binge Eating Disorder	Most common eating disorder in the U.S. Compulsive overeating without purging, accompanied by intense guilt and shame. Affects all body sizes; more common in males than any other eating disorder.

If you're concerned about yourself or someone you know: National Alliance for Eating Disorders Helpline: (866) 662-1235. Recovery is possible — early intervention dramatically improves outcomes.

Lesson 3 — Lifelong Nutrition

Vegetarian Diets

A **vegetarian** eats mostly or only plant-based foods. Well-planned vegetarian diets can be very healthy — lower in saturated fat, higher in fiber, and linked to reduced risk of cardiovascular disease and certain cancers. Key nutrients to watch: protein, iron, calcium, zinc, B vitamins, and Vitamin B12 (found almost exclusively in animal products — vegans should supplement).

Nutrition for Athletes

Teen athletes may need 2,000–5,000 calories/day. More carbohydrates for energy, more protein for muscle repair. Hydration is critical — drink before, during (every 15 minutes), and after exercise. **Making weight** by fasting or severe restriction harms both health and performance. The only healthy approach: gradual adjustment of ½–1 pound/week under guidance.

Performance Enhancers — Just Say No

- **Anabolic steroids** — Illegal without a prescription. Risk liver damage, stunted growth, heart problems, and permanently disrupted hormone development in teens.
- **Androstenedione** — Banned in professional sports. Similar risks to steroids with little proven benefit.
- **Creatine** — May give short power bursts; causes cramps, nausea, and at high doses can damage heart, liver, and kidneys.
- **Energy drinks** — High caffeine spikes heart rate; can cause dehydration, anxiety, and dangerous heart rhythm problems — especially in teens.

> Real performance gains come from consistent training, smart nutrition, quality sleep, and recovery — not from a pill or powder.

Chapter Vocabulary

Lesson 1

Metabolism	The process by which the body breaks down substances and gets energy from food
Body mass index (BMI)	A measure of body weight relative to height, used to assess healthy weight range
Overweight	Heavier than the standard weight range for one's height
Obese	Having an excess of body fat that carries significant health risks
Underweight	Below the standard weight range for one's height

Lesson 2

Body image	The way a person sees and feels about their own body
Fad diets	Weight-loss plans that tend to be popular for only a short time and typically don't produce lasting results
Weight cycling	A repeated pattern of losing and regaining body weight; also called 'yo-yo dieting'
Eating disorders	Extreme, harmful eating behaviors classified as mental illnesses that can cause serious physical damage or death
Anorexia nervosa	An eating disorder characterized by an intense fear of weight gain and severe restriction of food intake
Bulimia nervosa	An eating disorder involving cycles of binge eating followed by purging
Binge eating disorder	An eating disorder in which people compulsively overeat large amounts of food without purging

Lesson 3

Vegetarian	A person who eats mostly or only plant-based foods
Dietary supplements	Products that supply one or more nutrients as a supplement to a healthy diet
Performance enhancers	Substances used to boost athletic ability, most of which carry serious health risks
Herbal supplements	Dietary supplements made from plant extracts
Megadoses	Very large amounts of a supplement or vitamin, often dangerously exceeding recommended daily values